OTT CONTENT ANALYSIS DASHBOARD

PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF **BACHELOR OF ENGINEERING** IN **COMPUTER SCIENCE AND ENGINEERING** OF THE ANNA UNIVERSITY

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Submitted for the **Autonomous Semester End Mini Project Viva-Voce** held on

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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	TABLE OF CONTENTS		
CHAPTER NO.	**TITLE**		**PAGE NO.**
	ABSTRACT		1
	LIST OF FIGURES		2
1	**INTRODUCTION**		
	1.1	OBJECTIVES	3
	1.2	SCOPE OF THE PROJECT	3
2	**SYSTEM ANALYSIS AND SPECIFICATION**		
	2.1	PROBLEM DESCRIPTION	4
	2.2	FUNCTIONAL REQUIREMENT	5
	2.3	NON – FUNCTIONAL REQUIREMENT	6
3	**APPLICATION**		8
4	**OUTPUT AND RESULT**		9
5	**CONCLUSION & FUTURE ENHANCEMENT**		12
6	**REFERENCES**		13

ABSTRACT

The OTT Content Analysis Dashboard is a Business Intelligence (BI) project designed to analyze, visualize, and interpret content data from major streaming platforms such as Netflix, Amazon Prime Video, Hotstar, and Apple TV+. The rise of Over-The-Top (OTT) platforms has led to an explosion of digital entertainment data, making it challenging to evaluate content trends, audience preferences, and platform diversity. This project leverages Power BI to integrate and analyze multi-platform datasets, offering a unified analytical view of the OTT landscape.

The dashboard provides key insights through interactive visualizations, including total titles, genres, ratings, release trends, and type distribution (movies vs. TV shows). It enables users to compare platforms based on parameters like genre diversity, content volume, viewer ratings, and release year trends. With this analysis, stakeholders can identify popular genres, assess content growth over time, and understand platform-specific strengths.

By transforming raw OTT datasets into actionable insights, this BI solution enhances data-driven decision-making for media analysts, production houses, and marketers. The project demonstrates the power of Power BI in delivering visually engaging analytics that simplify complex entertainment data, helping organizations stay competitive in the rapidly evolving digital streaming industry.

LIST OF FIGURES

S.No	LIST OF FIGURES	PAGE NO
1	ott	9
2	hotstar	9
3	netflix	10
4	amazon prime	10
5	apple tv	11

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVES

The primary objective of this project is to design and implement a Business Intelligence (BI) system that provides a comprehensive analytical view of content across major Over-The-Top (OTT) platforms such as Netflix, Amazon Prime Video, Hotstar, and Apple TV+. By collecting and integrating data from various OTT sources—covering attributes like genres, release years, content types, and audience ratings—the system aims to simplify content comparison and offer real-time, actionable insights into platform performance and viewer preferences.

Additionally, The system enables data-driven decision-making for media analysts, production companies, and marketers. Through interactive Power BI dashboards, users can track KPIs like total titles, genre distribution, content type ratios, country-wise counts, and yearly release trends. It supports comparative and trend analysis to identify popular genres, growth patterns, and audience behaviors, helping drive strategic content planning and insights in the OTT industry.

1.2 SCOPE OF THE PROJECT

The scope of this project covers the collection, integration, analysis, and visualization of content data from multiple OTT platforms such as Netflix, Amazon Prime Video, Hotstar, and Apple TV+. It focuses on consolidating diverse datasets—covering attributes like genre, content type, ratings, release year, country, and cast—into a unified analytical platform. The system provides interactive Power BI dashboards that enable users, including analysts, content creators, and marketing teams, to explore and compare OTT content performance through dynamic visuals and intuitive filters.

Beyond basic reporting, the project scope extends to advanced analytical capabilities such as trend identification, genre-based performance analysis, and audience preference evaluation across platforms. This allows stakeholders to identify emerging content trends, assess platform diversity, and make data-driven strategic decisions. By leveraging Power BI's interactive features, the project ensures real-time insights, seamless navigation, and user-friendly data exploration without the technical expertise.

CHAPTER 2

SYSTEM ANALYSIS AND SPECIFICATION

2.1 PROBLEM DESCRIPTION

In today's digital entertainment era, vast amounts of data are generated every day across OTT platforms such as Netflix, Amazon Prime Video, Hotstar, and Apple TV+. However, this data often remains fragmented across multiple sources, making it difficult to derive meaningful insights about content performance, audience preferences, and platform growth. The absence of a centralized analytical system limits the ability of media analysts, production companies, and marketers to evaluate trends effectively or make informed strategic decisions.

Traditional content evaluation methods are largely manual, time-consuming, and non-interactive, which hinders comprehensive analysis of large datasets. Static reports fail to capture evolving content dynamics such as changes in viewer ratings, genre popularity, or release patterns. This makes it challenging for decision-makers to compare platforms, identify emerging trends, or understand audience engagement across different regions and content types.

Another major challenge lies in the lack of predictive and comparative analytics that can forecast future content demand or highlight performance differences among platforms. Without such analytical capabilities, OTT providers and analysts miss opportunities to optimize content strategies, marketing efforts, and investment decisions.

To address these challenges, the OTT Content Analysis Dashboard leverages Power BI to integrate, analyze, and visualize streaming data from multiple OTT platforms within a unified dashboard. The system provides interactive, real-time analytics through visual elements such as charts, filters, and KPIs—enabling users to explore total titles, genres, ratings, and release trends effortlessly. This BI solution transforms raw entertainment data into actionable insights, empowering stakeholders to make data-driven, proactive decisions and gain a competitive edge in the rapidly growing OTT industry.

2.2 FUNCTIONAL REQUIREMENT

1. **Data Integration and Import**
   - The system must allow importing of content-related data from multiple sources such as platform usage logs, user watch history, ratings, reviews, and subscriptions.
   - It should support standard file formats like CSV, Excel, or direct database/API connections for data loading.
2. **Data Processing and Transformation**
   - The system must clean, transform, and organize raw data to ensure consistency and accuracy.
   - It should enable mapping of content with unique identifiers (e.g., movie ID, series ID) to avoid duplication.
3. **Dashboard and Visualization**
   - The system must provide interactive dashboards for real-time visualization of content performance metrics.
   - It should display key indicators such as total views, average watch time, user engagement, ratings distribution, and trending content.
4. **User Role Management**
   - The system should allow different types of users (Admin, Content Manager, Marketing Analyst, Executive) with role-based access.
   - Each user should be able to view dashboards and reports relevant to their role.
5. **Performance Analysis and Reporting**
   - The system must generate automated reports on content popularity, user engagement, subscription trends, and regional preferences.
   - It should allow users to download or share reports easily for decision-making or strategic planning.
6. **Predictive Insights and Alerts**
   - The system should identify trending genres, at-risk content (low engagement), and potential content opportunities based on historical patterns.
   - It must provide recommendations for content promotion, retention strategies, or personalized user suggestions.
7. **Drill-down and Filtering Features**
   - The system must allow users to drill down into specific content titles, genres,

regions, or time periods.

- Users should be able to filter dashboards based on date ranges, subscription plans, device types, or demographic segments.

8. **Security and Data Privacy**
   - The system must ensure secure access to sensitive user and content data.
   - Only authorized users should be able to view or modify restricted information, in compliance with privacy regulations.

2.3 NON-FUNCTIONAL REQUIREMENT

1. **Performance**
   - The system should generate and load dashboards within a few seconds, even when handling large datasets from multiple OTT platforms.
   - Data refresh and visualization updates must occur smoothly without performance lag.
   - The system should support concurrent access by multiple users without affecting response time.
2. **Scalability**
   - The system must handle an increasing number of users, content titles, genres, and regions as the platform grows.
   - It should allow easy integration with additional data sources (e.g., new OTT platforms, user analytics tools) and reporting modules in the future.
3. **Security**
   - User and content data must be protected with secure access control, authentication, and encryption.
   - Only authorized users (admins, content managers, marketing analysts) should be able to access or modify data relevant to their roles.
   - Data sharing must comply with privacy regulations such as GDPR or platform-specific policies.
4. **Usability**
   - Dashboards should be user-friendly and easy to navigate for users with minimal technical knowledge.
   - Visuals should be clear, interactive, and support drill-down functionality for deeper insights.
   - Minimal training should be required for users to operate the system effectively.
5. **Reliability and Availability**
   - The system should be available at all times with minimal downtime.

- Dashboards and reports must provide accurate and consistent information at every refresh cycle.
- Backup and recovery mechanisms should be in place to avoid data loss.

6. Maintainability

- The system must be easy to update, maintain, and enhance with minimal disruption.
- New reports, visuals, or data sources should be easy to integrate without major architectural changes.

7. Portability

- The system should be accessible from different devices such as desktops, laptops, tablets, or smartphones.
- It should be compatible with multiple operating systems and modern web browsers for dashboard access.

CHAPTER 3

APPLICATION

1. **Content Performance Monitoring**
   - Helps streaming platforms track key metrics like view count, watch time, and audience ratings for each title.

2. **Genre and Trend Analysis**
   - Identifies popular genres, release trends, and viewer preferences across different time periods and regions.

3. **Audience Insights**
   - Analyzes user demographics, behavior patterns, and engagement levels to support personalized content recommendations.

4. **Market and Competitor Analysis**
   - Enables comparison of content distribution, popularity, and performance trends across multiple OTT platforms.

5. **Strategic Content Planning**
   - Assists media analysts and producers in making data-driven decisions about upcoming content, acquisitions, and promotions.

6. **Revenue and Subscription Insights**
   - Supports analysis of subscription growth, retention rates, and revenue contribution by content type or region.

7. **Decision Support for Marketing Teams**
   - Helps marketing teams identify high-performing content for targeted campaigns and promotional strategies.

8. **Operational Efficiency**
   - Reduces manual reporting by automating data visualization and providing real-time insights for faster decision-making.

CHAPTER 4

OUTPUT AND RESULT:

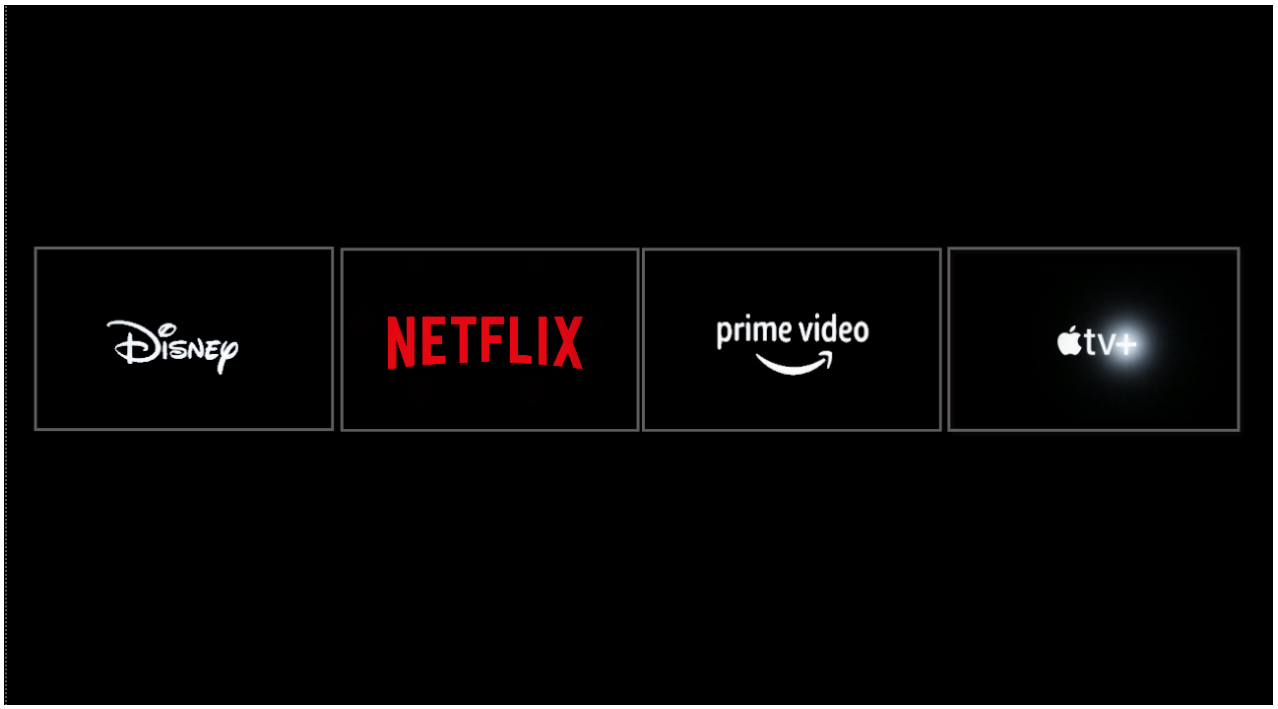

Figure 1: ott

Figure 2: hotstar

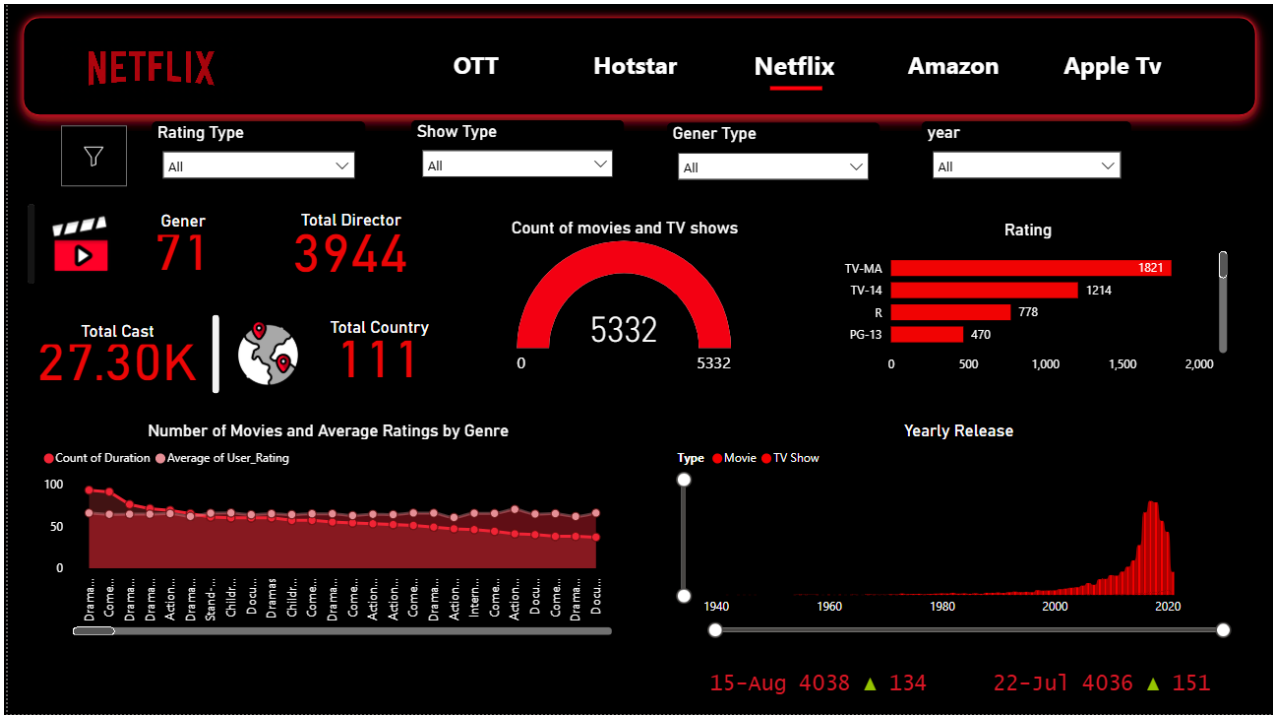

Figure 3: netflix

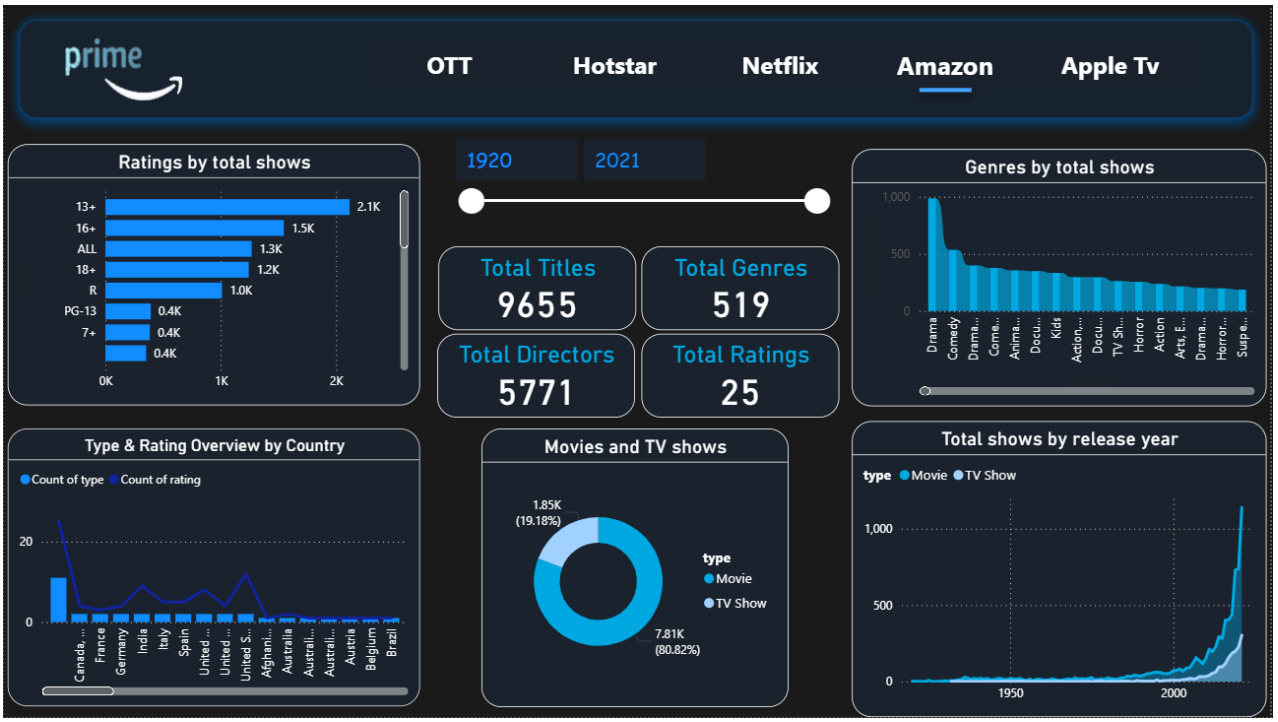

Figure 4: amazon prime

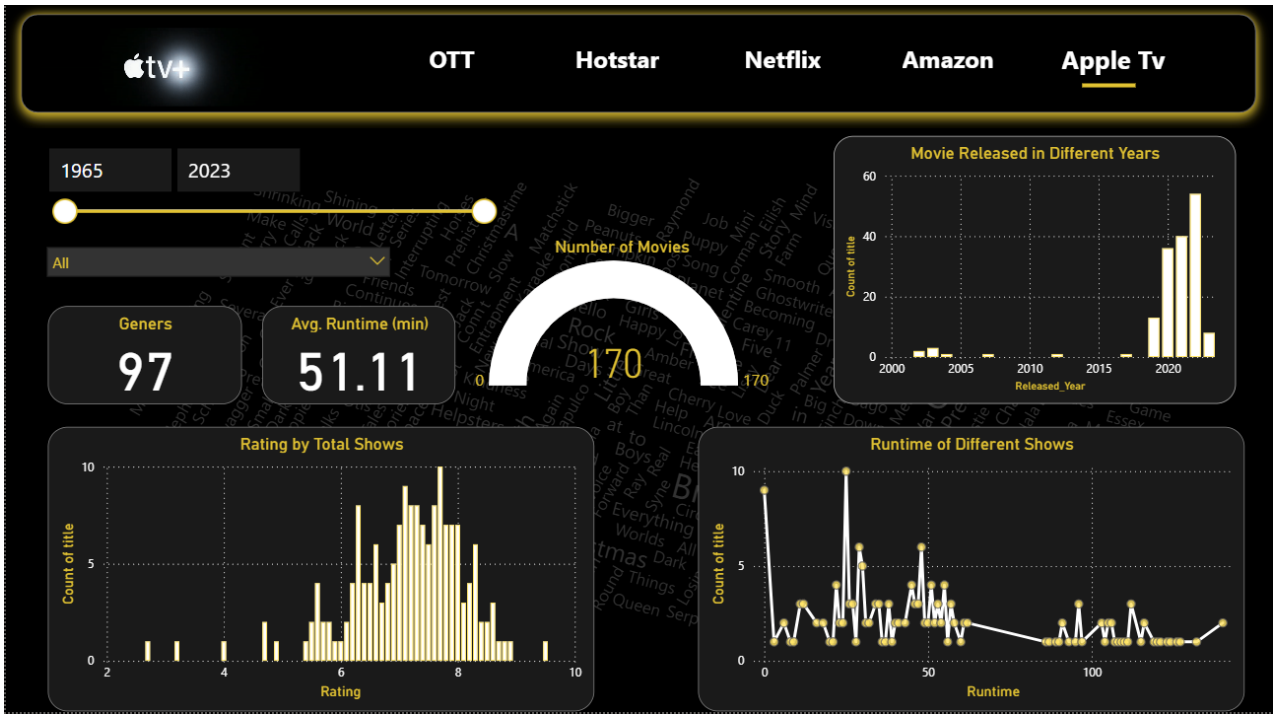

Figure 5: apple tv

OUTCOME

The OTT Content Analysis Dashboard effectively visualizes and compares content data from major streaming platforms — Hotstar, Netflix, Amazon Prime, and Apple TV. By analyzing key performance indicators such as genre distribution, content type ratio, country-wise content count, and yearly release trends, the dashboard provides meaningful insights into platform-specific strategies.

- Netflix shows a diverse global content base with a strong focus on Drama and Thriller genres.
- Amazon Prime Video highlights balanced growth across Action, Comedy, and Documentary content.
- Hotstar emphasizes regional and sports-related content, catering to Indian audiences.
- Apple TV focuses on premium, limited original productions with consistent quality ratings.

Overall, the dashboard demonstrates how Power BI-driven analytics can uncover platform trends, audience preferences, and content strategies—supporting data-driven decision-making in the OTT industry.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

The OTT Content Analysis Dashboard serves as an intelligent solution for monitoring and optimizing content performance across streaming platforms through data-driven insights. By integrating data from multiple sources—such as user watch history, ratings, reviews, subscriptions, and engagement metrics—the system provides a centralized platform for content performance tracking and audience analysis. With interactive dashboards and visualizations, stakeholders like content managers, marketing analysts, and executives can identify trending content, understand viewer preferences, and make informed decisions to improve content strategy. This enhances operational efficiency, promotes user engagement, and supports data-driven growth strategies for the OTT platform.

The project demonstrates how Business Intelligence can transform large volumes of streaming and user data into actionable insights, enabling better content planning, personalized recommendations, and targeted promotions. It simplifies reporting, improves transparency, and allows stakeholders to respond quickly to changing viewer behaviors and market trends.

Looking ahead, the system has strong potential for enhancements. Future developments could include real-time data streaming from OTT platforms for instant monitoring, integration of machine learning algorithms for content recommendation and predictive trend analysis, and support for personalized dashboards tailored to specific user segments. Additional improvements such as mobile application access, automated alerts for trending or underperforming content, and integration with marketing automation tools can further increase the system's functionality. These enhancements will elevate the dashboard from a content monitoring tool to a comprehensive OTT intelligence platform, driving data-informed decisions, user engagement, and platform growth.

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